Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/30/2022
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Project Title:	Identifying Genes Responsible for Adaptation to Spaceflight in a Photosynthetic Bacterium Using a Genome-Wide Fitness Screen		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:			
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	92093-0116	**Congressional District:**	52
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Task Description:	POSTDOCTORAL FELLOWSHIP Objectives: The goal of this project is to use advanced molecular biological tools to discover genes in a photosynthetic microorganism that are critical for adaptation to a spaceflight environment. Our objective is to compare growth on the International Space Station (ISS) to that of Earth and assess the survivability and adaptability of one of Earth's most wide-spread and robust photosynthetic organisms, cyanobacteria. Cyanobacteria are photosynthetic bacteria that fix carbon dioxide and generate oxygen from sunlight and have been exploited as an efficient biotechnology platform for synthesis of complex natural products, proteins, and lipids. These properties make them an ideal candidate for supporting human space exploration. However, few studies have been done assessing their potential for growth in space. Methodology: We will identify genes related to adaptation to spaceflight conditions using an existing state-of-the-art random-barcoded transposon (RB-TnSeq) library in the genetic model cyanobacterium Synechococcus elongatus (strain PCC 7942). Our lab has developed this genomic mutagenesis tool to quantitatively determine the fitness value of every gene in this organism under a wide variety of selective pressures. The experimental design of the proposed project is to grow the library under control and ground-based conditions of simulated microgravity and shortened photoperiod, and then to deep sequence the barcodes of the surviving population to determine the fitness of all genes in the organism. Bioinformatics is used to characterize the affected genes and metabolic pathways. This is a powerful method to identify genes providing survival benefits or liabilities in space. These experiments will give us a genome-wide look at which genes are helpful and which harmful for adapting to space. Genes identified during the initial screen will be subjected to further site-specific mutation and analysis, and engineered mutants will be constructed for further testing to assess adaptation to spaceflight. These studies will be supported by our CYANO-VECTOR gene assembly kit, which allows for quick and customizable gene expression studies in a diverse set of cyanobacterial strains. This project will provide a comprehensive analysis of genes important during spaceflight in a photoautotroph and lay the foundation for genetically engineering organisms to survive and thrive during extra-planetary trips. Significance: The proposed project meets ROSBio 2016 call priorities "1) effectively use microgravity and other space environment characteristics to enhance our understanding of the adaptation and function of basic biological processes in spaceflight and 2) develop a scientific and technological knowledge base that will contribute to a safe, productive human presence in space." This study is ideal for Element 3.4: Molecular and Cellular Biology, Focus c: "Investigations that characterize the role of a gene, or a set of genes, in an organism's ability to acclimate to spaceflight." This project uses cutting-edge molecular biology tools and techniques in a model photoautotroph to perform an important screen of all possible gene effects. Previous studies on a diverse set of organisms have shown that spaceflight affects cellular oxidative stress response pathways, cell wall formation, and cell morphology. This project will help create a roadmap for purposeful genetic engineering to generate more robust and adaptable organisms. Cyanobacteria are an under-utilized resource for supporting human space exploration. They could serve as a vital component of biological life support systems: generating oxygen from CO2, H2O, and light; recycling nutrients; and synthesizing food and nutritional compounds such as omega-3 fatty acids. This study would lay the foundation for the use and adaptation of cyanobacteria to address the challenges living organisms face in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has benefits for both space travel and the Earth, and the potential for future impact is high. Cyanobacteria are some of the most important primary producers on Earth, producing half of our oxygen, and capturing CO2 from the atmosphere by photosynthesis. Further study on how these organisms cope with environmental stress and adapt to changing conditions may have important implications for climate change on Earth as well as space travel. The development of sustainable algae culture conditions in space could have direct benefits for Earth. Cyanobacteria can be thought of as tiny biological carbon capture devices that run on sunlight and can be grown on wastewater, and they could prove invaluable for sustainable resource production in space as well as on Earth.
	Main goals and objectives The main objective of this project is to study photosynthetic cyanobacteria under ground-based conditions mimicking those of Low-Earth Orbit (LEO) with the goal of advancing the use of cyanobacteria in bioregenerative life support (BLS) systems. Cyanobacteria, a large family of photoautotrophic bacteria, are well qualified to perform key functions to support human space exploration. Fossils composed of cyanobacteria are some of the oldest known records of life on Earth, and cyanobacteria are the only organisms to have evolved oxygenic photosynthesis. Through oxygenic photosynthesis, cyanobacteria both remove CO2 from the air and generate oxygen, while efficiently harvesting sunlight energy, and are some of the most important primary producers on the planet. Many strains are safe for consumption by animals, including humans. Cyanobacteria could therefore play a significant role in BLS for future space missions, as they require little in the way of nutrients, and could even be used for wastewater recycling. Studying how these microorganisms respond and adapt to spaceflight will therefore be important for establishing a sustainable space-based human presence. Our experiments focus on the adaptations of cyanobacteria to ambient light conditions during spaceflight in LEO. One energy-saving way to culture cyanobacteria in space would be to grow them using ambient sunlight, which could be complicated by the light cycle of the International Space Station (ISS), which orbits Earth roughly once every 90 minutes, rapidly moving in and out of sunlight 16 times over 24 hours with a 45-minute light and 45-minute dark (45':45' LD) period. This period of lighting is highly unusual for organisms that evolved with a consistent day-night cycle, and that maintain a molecular clock with a roughly 24-hour period that regulates cellular metabolism and activity. One of the primary goals was to identify whether cyanobacteria would tolerate these lighting conditions, and to determine genes that could impact their adaptation to such conditions. To examine growth under 45':45' LD, we selected three lab-cultivated strains of cyanobacteria, Synechococcus elongatus PCC 7942 (hereafter S. elongatus), Anabaena PCC 7120 (hereafter Anabaena), and Leptolyngbya BL0902 (hereafter Leptolyngbya). These strains were chosen for their diverse characteristics and for their genetic tractability with established genetic methods and tools for engineering genome alterations. In addition, we used three green algal strains, Scenedesmus dimorphus UTEX1237 (hereafter Scenedesmus), Chlamydomonas reinhardtii CC-5082 (hereafter Chlamydomonas), and Chlorella vulgaris UTEX395 (hereafter Chlorella). All strains were grown under one of three

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light conditions, with continuous light at half intensity, or 45':45' LD and 12h:12h LD at full intensity. Over 24 h, each strain received the same amount of light delivered in different light periods. Each cyanobacterial strain grew the fastest when supplied with continuous light, grew the slowest in 12h:12h LD, and grew at an intermediate rate under 45':45' LD. The green algal strains again grew fastest under continuous light, but there was little difference between 12h:12h LD and 45':45' LD. Importantly, all strains of green algae and cyanobacteria tested grew under continuous light and the unusual 45':45' LD regime, demonstrating a wide range of light period tolerances, and ability to adapt to the conditions of LEO.

After our initial experiments showed that cyanobacteria grow well in 45':45' LD, we decided to focus on the strain S. elongatus, which has well developed genetic tools and an established circadian clock. Circadian clocks are ancient mechanisms that help to synchronize cellular activities with the Earth's 24 h day-night cycle. We monitored circadian rhythms by measuring bioluminescence from the AMC541 reporter strain in which the luc gene encoding luciferase is driven by a promoter that oscillates its activity throughout the 24 h cycle and produces rhythmic bioluminescence from the cells. We entrained two plates to 12h:12h LD in opposite phases and then moved them into a single 12h:12h LD cycle, 45':45' LD cycle, or continuous light, and monitored the bioluminescence every 90 min for 7 days. Growth in a 12h:12h cycle quickly entrained both plates to the same synchronization. In continuous light and 45':45' LD, however, the plates maintained their opposite-phase entrainment throughout the experiment. There was also little difference in the free-running period, the period at which the clock oscillates in the absence of light-dark signals. These data show that the 45':45' LD periods don't affect the normal cycling of the internal circadian clock, and that growth under 45':45' LD more closely resembles that of continuous light than 12h:12h LD.

We used a randomly barcoded transposon insertion-site sequencing (RB-TnSeq) mutant library in S. elongatus to identify genes that affect the fitness of S. elongatus in 45':45' LD. The RB-TnSeq library contains cells with single transposon insertions at random locations in the genome, including regulatory and coding sequences. By growing the library under 45':45' LD, we can identify mutations in genes that are harmful or beneficial to growth under those conditions. From this screen, we found several mutants that grew significantly more slowly in 45':45' LD than in continuous light. Five of our top genes are all involved in glycogen degradation/oxidative pentose phosphate pathway (OPPP) in S. elongatus. The OPPP is responsible for utilizing an energy storage molecule, glycogen, for energy used to fuel the cell's metabolism in the dark.

We engineered insertion mutants in each of five genes to determine the effect of the OPPP and glycogen metabolism in response to growth in 45':45' LD. When grown in continuous light, each of the mutants grew about as well as wild type (WT), showing no growth defect. However, when grown in 45':45' LD, each of the OPPP mutants grew significantly slower than WT. This demonstrates that in order to maintain robust growth in 45':45' LD, the cyanobacterial cell must be able to break down glycogen harvest chemical energy for the cell. Even for short dark periods of 45 min, the ability to generate energy in the dark significantly affects metabolism. We picked two genes to complement with WT alleles to confirm that the slower growth rate was tied directly to the knockout of the OPPP genes and not to second-site mutations. The complementation of these two genes was successful, and both complemented strains grew as well as WT in 45':45' LD while the knockout strains again grew more slowly. We also performed cell viability assays by fluorescence and light microscopy to determine whether the fitness effects were due to slower growth, or cell death. There was no difference in cell mortality between continuous light and 45':45' LD, which indicates that cells in 45':45' LD grow more slowly than in continuous light, but remain viable.

Conclusions

In these experiments we demonstrated that the constant 45-minute "day" and 45-minute "night" of LEO provides a suitable light environment for the growth of green algae and cyanobacteria. The 45':45' LD cycle appears to be too short a time to disrupt the rhythms of the cyanobacterial circadian clock that evolved with a 24 h day. Unexpectedly, all the WT cyanobacterial strains grew better in 45':45' LD than in 12h:12h, despite receiving the same amount of photon flux. The green algae strains tolerated 45':45' LD, but there was no increase in growth rate compared to 12h:12h LD.

Another important finding is that cellular energy storage and retrieval pathways such as the OPPP are important for continuous growth during short periods of darkness. The viability of OPPP mutants in 45':45' LD suggests that the LEO light exposure regimen does not generate toxic levels of reactive oxygen species during the light period. Together, these data show that the timing of the light and dark periods is an important factor for cyanobacterial growth, even in unusual regimes like 45':45' LD. Light exposure will be an important consideration when utilizing cyanobacteria in a BLS system in the space environment. For space journeys outside of Earth's orbit, or in the lunar polar region, sunlight will be nearly constant, an environment in which cyanobacteria and green algae thrive. LEO, the location of the ISS and planned future space stations, is an ideal place to test cyanobacteria-based BLS systems using ambient sunlight. Microgravity appears not to be an impediment to their growth as demonstrated by Poughon et al. for Limnospira indica. With short generation times, small genomes, and robust genetic tools like those used in this work, cyanobacteria are excellent candidate organisms for investigating bacterial adaptation to a space environment in support of human exploration.

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Bibliography Type:	Description: (Last Updated: 06/08/2023)
Articles in Peer-reviewed Journals	Bishé B, Golden SS, Golden JW. "Glycogen metabolism is required for optimal cyanobacterial growth in the rapid light-dark cycle of low-Earth orbit." Life Sci Space Res. 2023 Feb;36:18-26. https://doi.org/10.1016/j.lssr.2022.11.001 ; PMID: 36682825; PMCID: PMC9989776 , Feb-2023